Mongoose

Mongoose

- **Mongoose** is an **Object Data Modeling (ODM) library** for **MongoDB** and **Node.js**.
- It provides a **straightforward and schema-based** way to interact with MongoDB databases in Node.js applications.

Features

- **Schema-based Modeling**: Define how your documents should look using schemas.
- **Data Validation**: Enforces data types and rules (e.g., required fields).
- **Middleware Support**: Add logic before or after events (e.g., before saving a document).
- **Built-in Type Casting**: Automatically converts data types (e.g., strings to dates).
- **Query Building**: Simplifies complex MongoDB queries.
- **Easy CRUD operations**

Why Use Mongoose?

- While MongoDB is schema-less, Mongoose gives structure to your data.
- This is especially useful for building reliable and maintainable applications.

Example Use Case

- If you're building a student record system, you can define a Student schema with name, regNumber, and age, and Mongoose will handle **how the data is stored, validated, and retrieved from MongoDB.**

Connecting to MongoDB with Node.js (Without Mongoose)

Install mongodb driver:

- npm init -y
- npm install mongodb

Connecting to MongoDB with Node.js (Without Mongoose)

connectMongo.js

```
const { MongoClient } = require('mongodb');
// MongoDB local connection
const uri = 'mongodb://localhost:27017';
const client = new MongoClient(uri);

async function run() {
  try {
    await client.connect();
    console.log('Connected to MongoDB');

    const db = client.db('testDB');
    const collection = db.collection('students');
    const result = await collection.insertOne({ name: 'John', age: 21 });
    console.log('Inserted:', result.insertedId);
  } catch (err) {
    console.error(err);
  }
finally {
    await client.close();
  }
}

run();
```

- **To run: node connectMongo.js**

Connecting to MongoDB using Mongoose

To install mongoose:

Install it:

npm install mongoose

mongooseConnect.js

```
const mongoose = require('mongoose');


// Connect to local MongoDB
mongoose.connect('mongodb://localhost:27017/studentDB', {
  useNewUrlParser: true,
  useUnifiedTopology: true,
});


const db = mongoose.connection;


db.on('error', console.error.bind(console, 'Connection error:'));
db.once('open', () => {
  console.log('Connected to MongoDB via Mongoose');
});
```

Defining Mongoose Schemas

Schema: Defines the structure of a document.
Model: A constructor based on a schema to create and interact with documents.

studentModel.js

```
const mongoose = require('mongoose');

const studentSchema = new mongoose.Schema({
  name: {
    type: String,
    required: true,
  },
  registerNumber: {
    type: String,
    required: true,
    unique: true,
  },
  age: Number,
});


const Student = mongoose.model('Student', studentSchema);


module.exports = Student;
```

CRUD Operations using Mongoose

app1.js (Main app)

```
const mongoose = require('mongoose');
const Student = require('./studentModel');


// Connect to MongoDB
mongoose.connect('mongodb://localhost:27017/studentDB', {
  useNewUrlParser: true,
  useUnifiedTopology: true,
});
```

```
// Create (Insert)
async function createStudent() {
  const student = new Student({
    name: 'Alice',
    registerNumber: 'REG101',
    age: 22,
  });
  await student.save();
  console.log('Student Created:', student);
}
```

CRUD Operations using Mongoose contd...

```
// Read (Find)
async function getStudents() {
  const students = await Student.find();
  console.log('All Students:', students);
}

// Update
async function updateStudent(regNo) {
  const student = await Student.findOneAndUpdate(
    { registerNumber: regNo },
    { age: 23 },
    { new: true }
  );
  console.log('Student Updated:', student);
}
```

CRUD Operations using Mongoose contd...

```
// Delete
async function deleteStudent(regNo) {
  const result = await Student.deleteOne({ registerNumber: regNo });
  console.log('Deleted:', result);
}

// Call the functions here
async function main() {
  await createStudent();
  await getStudents();
  await updateStudent('REG101');
  await deleteStudent('REG101');
  mongoose.connection.close();
// Close the DB after operations
}

main();
```

To Run in VS Code:

node app.js